🧩 Detailed BRD: Point of Sale

Overview

The Simplified Point of Sale (POS) module is the main interface for hardware store staff. Since staff have limited technical skills, the module is distraction-free, intuitive, and free of complex features. Its main functions let staff complete transactions and trigger background processes (inventory deduction and transaction logging) without needing advanced computer knowledge.

Scope

- Browsing or searching for available hardware items
- Adding items and adjusting quantities in a virtual cart
- Real-time calculation of subtotals and grand totals
- Finalizing a checkout transaction
- Automated deduction of sold items from the central inventory
- Automated logging of the transaction (including timestamp and the active staff's username)

Out-of-Scope

- Direct hardware integration with physical credit card terminals (payments are assumed cash or handled on a separate manual terminal)
- Barcode scanner or receipt printer integration
- Complex returns, refunds, or exchange processing
- Applying discounts or coupons

Functional Requirements

Requirement	Actor	Priority	Rationale
The system shall allow the user to search for an item by its name or ID.	Staff	Must	Cashiers need a fast way to locate specific hardware items to add to the sale.

The system shall allow the user to add selected items to a current transaction (cart) and define the quantity.	Staff	Must	Core functionality for processing a customer's purchase.
The system shall automatically calculate and display the total price of all items in the current transaction.	System	Must	Eliminates manual math errors for the cashier.
The system shall prevent the user from adding a quantity of an item that exceeds the current available stock.	System	Must	Prevents selling items that are physically out of stock.
The system shall allow the user to click a single, prominent "Complete Sale" button to finalize the transaction.	Staff	Must	Meets the "super basic GUI" requirement for a frictionless checkout process.
Upon finalizing a sale, the system shall automatically deduct the sold quantities from the central inventory database.	System	Must	Keeps the Admin dashboard and demand forecasting logic accurate and up-to-date.
The system shall generate a permanent transaction record containing the items sold, total amount, timestamp, and the cashier's username.	System	Should	Fulfills the requirement to track who handled the transaction for employee accountability.

The system shall clear the cart and reset the POS interface to a blank state immediately after a sale is finalized.	System	Should	Prepares the terminal for the next customer in line without manual clearing.

Non-Functional Requirements

Category	Requirement
Usability	The POS interface shall utilize large typography, high-contrast colors, and a maximum of 4 interactive elements (Search, Add, Quantity Adjust, Checkout) on the main screen.
Performance	The total price calculation must update in under 0.5 seconds whenever an item quantity is adjusted.
Reliability	The "Complete Sale" button shall disable immediately after one click to prevent accidental double-billing or double-inventory deduction.
Usability	The interface must be fully operable on standard resolution desktop monitors without requiring vertical or horizontal scrolling for the main cart view.

Constraints

- The system requires an active local network or internet connection to communicate with the central database; there is no offline mode.
- Prices are strictly governed by the Admin; Staff cannot modify the unit price of an item during checkout.

Limitations

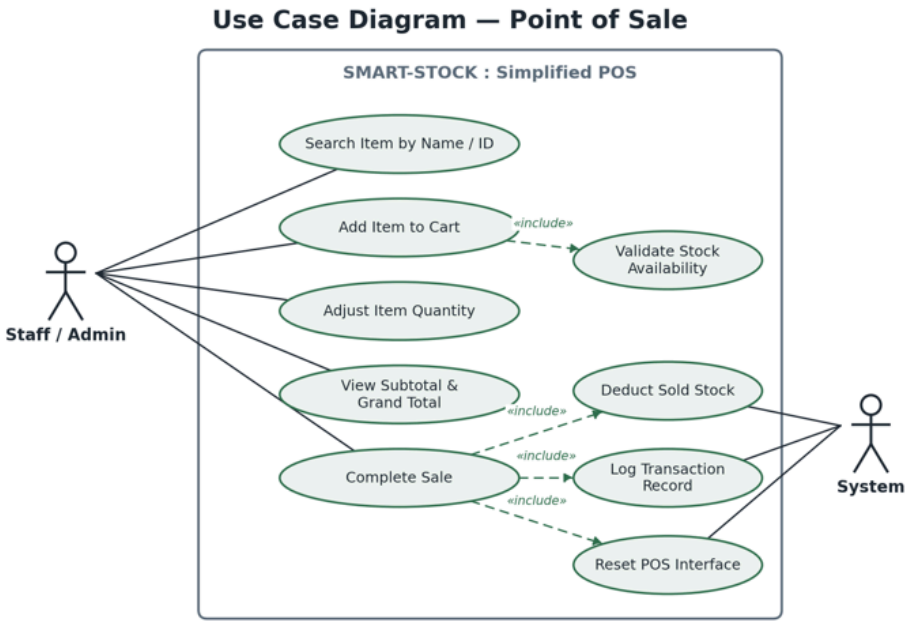
- Because there is no barcode scanner in this phase, search speed relies entirely on the staff's typing speed and the search algorithm's efficiency.

Assumptions

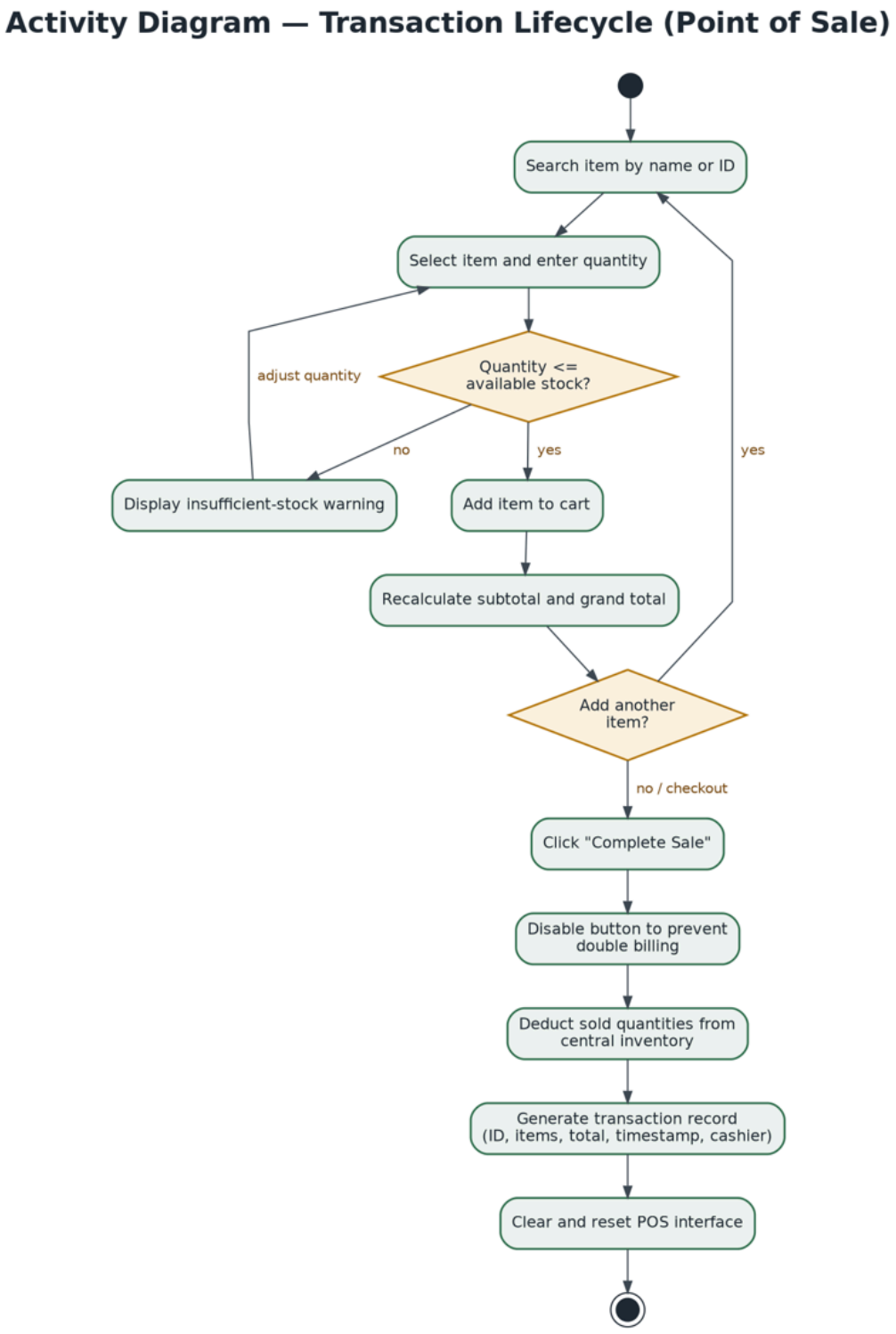
- All product details (name, price, initial stock) have been pre-configured in the database by the Admin prior to the Staff initiating a sale.
- Staff members will visually verify that the physical stock at the counter matches before clicking "Complete Sale."

Requirements Model

- **Use Case Diagram**



- **Business rules**
  - Cart Total = Sum of (Item Unit Price * Quantity).
  - Inventory Deduction = Current Stock - Sold Quantity.
  - Cannot checkout if Cart is empty or if Cart contains out-of-stock items.
- **Activity Diagram**
  - *Transaction Lifecycle*



- **State model**
  - *Cart State Lifecycle*

